PHYSICAL CHEMISTRY

GUIDED REVISION PLAN-SCORE ADVANCED

Topic : Atomic Structure (Question Bank # 12)

MM : 81 **Time : 45 Min.**

SECTION–I : (i) Only One option correct Type

This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. The wavelength of K_α-characteristic X-rays produced is λ, when cathode rays strike a metal of atomic number Z. What should be the atomic number of metal such that it can produce the K_α-characteristic X-rays of wavelength 4λ?

(A) $\frac{Z}{16}$ (B) $\frac{Z}{2}$

(C) $\frac{Z+1}{2}$ (D) $2Z - 1$

2. According to Maxwell's theory of electrodynamics, an electron going in a circle should emit radiation of frequency equal to its frequency of revolution. What should be the wavelength of the radiation emitted by a hydrogen atom in the ground state if this rule is followed?

(A) 4500 nm (B) 450 nm

(C) 45 nm (D) 4.5 nm

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3. An electron revolving around H-nucleus in ground state absorbs 10.2 eV energy. Its angular momentum increases by

(A) $\frac{h}{2\pi}$ (B) $\frac{h}{\pi}$

(C) $\frac{2h}{\pi}$ (D) $\frac{h}{4\pi}$

4. In a hydrogen like species , an electron jumps from an orbital having two radial and two angular nodes to the orbital having same sign of wave function in all direction, at any distance. If the energy of emitted photon is 326.4 eV, the atom or ion is

(A) H (B) Li^{2+}

(C) He^{+} (D) B^{4+}

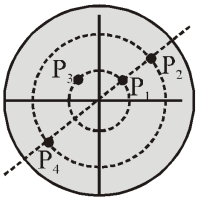
5. O_2 undergoes photochemical dissociation into one normal oxygen atom and one oxygen atom, 1.2 eV more energetic than normal. The dissociation of O_2 into two normal atoms of oxygen requires 482.5 kJ/mol. The maximum wavelength effective for photochemical dissociation of O_2 is (1 eV = 96.5 kJ/mol)

(A) 248 nm (B) 1033.3 nm (C) 1236.2 nm (D) 200 nm

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6. The given diagram shows points P_1, P_2 P_3 & P_4 in 1s orbital.



1s orbital

The correct order of radial probability density at these point are

(A) $P_1 = P_2 = P_4 > P_3$

(B) $P_1 = P_3 < P_2 = P_4$

(C) $P_1 = P_3 > P_2 = P_4$

(D) $P_1 = P_2 = P_4 < P_3$

7. Which electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon

(A) 3s (B) 2p

(C) 2s (D) 1s

8. Assuming Heisenberg Uncertainity Principle to be true what could be the minimum uncertainty in de-broglie wavelength of a moving electron accelerated by Potential Difference of 6 V whose uncertainty in position is $\frac{7}{22}$ n.m.

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(A) 6.25 Å (B) 6 Å

(C) 0.625 Å (D) 0.3125 Å

9. A light source of wavelength λ illuminates a metal and ejects photo-electrons with $(KE)_{max}$ = 1 eV. Another light source of wavelength, $\frac{\lambda}{3}$, ejects photo-electrons from same metal with $(KE)_{max}$ = 4 eV. Find the value of work function ?

(A) 1 eV (B) 2 eV

(C) 0.5 eV (D) None of these

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(ii) One or more options correct Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

10. The orbital angular momentum of an electron in a single electron system is $\sqrt{3}\frac{h}{\pi}$. Which of the following angular momentum value (s) are not possible for this electron in Bohr orbit.

(A) $\frac{3h}{2\pi}$ (B) $\frac{h}{\pi}$

(C) $\frac{3h}{\pi}$ (D) $\frac{2h}{\pi}$

11 For He^+ ion, select incorrect option(s) :

(A) Difference between radius of II & III Bohr's orbit is greater than that of III & IV Bohr orbit.

(B) Ratio of velocity of electron in III & IV Bohr's orbit is less then one.

(C) Energy difference between consecutive energy levels decreases continuously.

(D) Ratio of wavelength of least energetic and most energetic photon of lyman series is (3/4).

12. Which of the following orbitals has (have) two spherical nodes?

(A) 2s (B) 4p

(C) 3d (D) 6 f

13. Orbitals in which the numbers of radial and angular nodes are equals?

(A) 3p (B) 4d

(C) 5d (D) 6f

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14. When photons of energy 4.25 eV strike the surface of a metal 'A', the ejected photo electrons if maximum kinetic energy have wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal 'B' by photons of energy 4.7 eV is 1.5 eV less than that of photo electrons of metal A. If wavelength of photo electrons of metal B is twice than that of photoelectron of 'A' then -

(A) Work function of A = 2.25 eV

(B) Work function of B = 4.2 eV

(C) Maximum (K.E.) of photoelectrons of 'A' = 2 eV

(D) Maximum (K.E.) of photoelectrons of 'B' = 2.75 eV

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(iii) Paragraph Type

This section contains **2 paragraphs** , describing theory, experiment, data etc. **6 questions** relate to 2 paragraphs with 3 questions. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for questions 15 to 17

The hydrogen-like species Li^{2+} is in a spherically symmetric state S_1 with one radial node. Upon absorbing light the ion undergoes transition to a state S_2. The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom.

15. The state S_1 is :-

(A) 1s (B) 2s (C) 2p (D) 3s

16. Energy of the state S_1 in units of the hydrogen atom ground state energy is :-

(A) 0.75 (B) 1.50 (C) 2.25 (D) 4.50

17. The orbital angular momentum quantum number of the state S_2 is :-

(A) 0 (B) 1 (C) 2 (D) 3

Paragraph for question nos. 18 to 20

The location of electron inside the atom can be predicted by wave mechanical model which gives wavefunction of orbitals represented as $\Psi_{n,\ell,m}$ where n,ℓ, m represents quantum numbers sometimes, Ψ can be analysed as two separate components , radial and angular, represented as $\Psi_{(r)}$ and $\Psi_{(\theta,\phi)}$ respectively for H, the following expressions of $\Psi_{n,\ell,m}$ are known

[Where a_0 is constant and σ is $\frac{2r}{na_0}$]

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$$\Psi_{1,0,0} = \frac{2}{a_0^{3/2}} e^{-\sigma/2} \qquad \Psi_{3,0,0} = \frac{1}{9\sqrt{3}}\left(\frac{1}{a_0}\right)^{3/2}\left(6-6\sigma+\sigma^2\right)e^{-\sigma/2}$$

$$\Psi_{2,0,0} = \frac{1}{2\sqrt{2}}\left(\frac{1}{a_0}\right)^{3/2}(2-\sigma)e^{-\sigma/2} \qquad \Psi_{3,1,0} = \frac{1}{9\sqrt{3}}\left(\frac{1}{a_0}\right)^{3/2}(4-\sigma)\sigma e^{-\sigma/2}$$

18. For 1 s orbital H-atom, the radial probability will be maximum at what radial distance from nucleus:

(A) r = 0 (at the nucleus) (B) r = ∞

(C) r = a_0 (D) r = $2a_0$

19. At what radial distance will there be a spherical node for $3p_y$ orbital of hydrogen?

(A) r = $6a_0$ (B) r = $2a_0$ (C) r = 0 (D) data is insufficient.

20. Which of the following statements are incorrect w.r.t. $\Psi_{(r)}$ v/s r graph for H-atom

(A) For 1s orbital the graph will not intersect x - axis at finite distance.

(B) For 2s orbital, the graph will have one 'local minima' only.

(C) For 3s orbital, the graph will intersect x - axis at two distinct points.

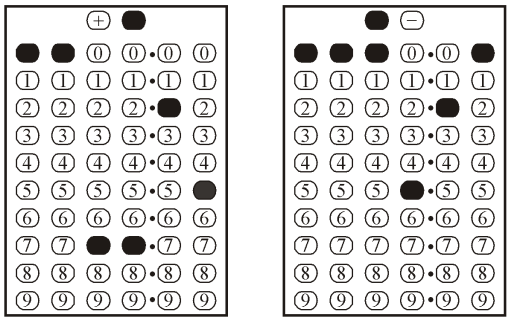
(D) For 3p orbital, the graph will intersect x-axis at two non-zero distinct points.

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SECTION-II: (Integer Second Decimal)

- This section contains **3** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. How many non-spherical orbitals are possible that have atleast two maxima if a curve is plotted between radial probability distribution function versus radial distance for which principal quantum number: $n \leq 4$?

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2. If 4 times the most probable distance of electron of a 1s orbital in a unielectronic atom /ion is given by 0.529 Å then the atomic number of the unielectronic species will be.

[Given : Wave function of 1s orbital, $\Psi = \sqrt{\left(\frac{Z^3}{\pi a_0^3}\right)} e^{-Zr/a_0}$ where a_0 = radius of first Bohr's orbital in $H-atom \equiv (52.9pm)$ and Z is atomic number of unielectrnic species]

3. Find the number of spectral lines in Paschen series emitted by atomic H, when electron is excited from ground state to 7th energy level returns back.

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SECTION-IV : (Matrix Type)

This Section contains **1 question**. Question has **four statements** (A, B, C and D) given in **Column I** and five statements (P, Q, R, S and T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1. Match the column -

Column-I		**Column-II**	
(A)	l	(P)	Indicates the distance from nucleus
(B)	s	(Q)	It can have only two possible values.
(C)	m	(R)	Possible values are 0 to (n – 1)
(D)	n	(S)	Indicates the possible orientations in space.
		(T)	Possible values are $+l$ to $-l$ including 'Zero'

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ANSWER KEY

SECTION–I : (i) Only One option correct Type 3(–1)

1. Ans.(C)	2. Ans.(C)	3. Ans.(A)	4. Ans.(D)
5. Ans.(D)	6. Ans.(C)	7. Ans.(D)	8. Ans.(C)
9. Ans.(C)			

(ii) One or more options correct Type 4(–1)

10. Ans.(A,B)	11. Ans.(A,B,D)	12. Ans.(B,D)	13. Ans.(A,C)
14. Ans.(A,B,C)			

(iii) Paragraph Type 3(0)

15. Ans.(B)	16. Ans.(C)	17. Ans.(B)	18. Ans.(C)
19. Ans.(C)	20. Ans.(D)		

SECTION-II: (Integer Second Decimal)

1. Ans.(11.00)	2. Ans.(4.00)	3. Ans.(4.00)

SECTION-IV : (Matrix Type)

1. Ans(A) - R ; (B) - Q ; (C) - S, T ; (D) - P

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